

Good attempt
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National University of Computer and Emerging Sciences
FAST School of Computing Fall-2022 Islamabad Campus

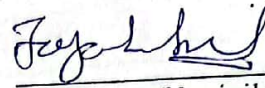
MT-1003: Calculus and
Analytical Geometry (CS)

Serial No:
Sessional Exam-I
Total Time: 1 Hour
Total Marks: 60

Tuesday, 27th September, 2022

Course Instructors

Dr. Hamda Khan, Dr. Imran Shahzad, Dr. Naveed
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Signature of Invigilator

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Student Name Roll No. Section


Signature

DO NOT OPEN THE QUESTION BOOK OR START UNTIL INSTRUCTED.

Instructions:

1. Attempt on question paper. Attempt all of them. Read the question carefully, understand the question, and then attempt it.
2. No additional sheet will be provided for rough work. Use the back of the last page for rough work.
3. If you need more space, write on the back side of the paper and clearly mark question and part number etc.
4. After asked to commence the exam, please verify that you have seven (07) different printed pages including this title page. There are a total of 5 questions.
5. Calculator sharing is strictly prohibited.
6. Use permanent ink pens only. Any part done using soft pencil will not be marked and cannot be claimed for rechecking.
7. Solve the questions using techniques learnt in this course.

	Q-1	Q-2	Q-3	Q-4	Q-5	Total
Marks Obtained	14	18	9	10	07	50
Total Marks	20	10	10	10	10	60

+3.5 = 53.5

Question 1A [10 Marks]

A piece of wire 12 inches long is to be used to form a square and/or a circle. Determine a function that expresses the combined area of both figures. What is its domain?

For Square:

$$\text{Perimeter} = 4u$$

$$12 = 4u \Rightarrow \boxed{u = 3}$$

For Circle:

$$\text{Circumference} = 2\pi r$$

$$12 = 2\pi r \Rightarrow \pi r = 6 \Rightarrow \boxed{r = \frac{6}{\pi}}$$

Area of Square:

$$A_1 = u^2 \Rightarrow \boxed{A_1 = 9}$$

Area of Circle:

$$A_2 = \pi r^2 \Rightarrow A_2 = \pi \cdot \frac{36}{\pi^2} \Rightarrow \boxed{A_2 = \frac{36}{\pi}}$$

Combined Area

$$A = A_1 + A_2$$

$$A = u^2 + \pi r^2 \quad \text{--- (1)}$$

Now,

$$12 = 4u + 2\pi r \quad \checkmark$$

$$6 = 2u + \pi r$$

$$\pi r = 6 - 2u \Rightarrow \boxed{r = \frac{6 - 2u}{\pi}} \quad \checkmark$$

~~A~~

$$A(u) = u^2 + \pi \left(\frac{6-2u}{\pi} \right)^2$$

$$A(u) = u^2 + \pi \frac{(6-2u)^2}{\pi^2}$$

$$A(u) = u^2 + \frac{(6-2u)^2}{\pi} \quad \text{--- (i)}$$

$$u^2 + 36 + 4u^2 - 24u + 36$$

$$3u^2 - 24u + 36$$

Also,

$$6 = 2u + \pi x$$

$$6 - \pi x = 2u$$

$$u = \frac{6 - \pi x}{2}$$

~~$$A(x) = 2 \left(\frac{6 - \pi x}{2} \right)^2$$~~

$$A(x) = \left(\frac{6 - \pi x}{2} \right)^2 + \pi x^2$$

$$A(x) = \frac{(6 - \pi x)^2}{4} + \pi x^2 \quad \text{--- (ii)}$$

Dom ??

07.5

$$36 + \pi x^2 - 12\pi x + \pi x^2$$

Question 1B [10 Marks]

An aircraft manufacturer wants to determine the best-selling price for a new airplane. The company estimates that the initial cost of designing the airplane and setting up the factories in which to build it will be 500 million dollars. The additional cost of manufacturing each plane can be modeled by the function:

$$m(x) = 20x - 5x^{3/4} + 0.01x^2.$$

where x is the number of aircraft produced and m is the manufacturing cost, in millions of dollars. The company estimates that if it charges a price p (in millions of dollars) for each plane, it will be able to sell $x(p) = 320 - 7.7p$ planes.

- Find the cost, demand, revenue, and profit functions.
- Find the marginal profit at $x = 12$, and comment on the results.

Fixed cost = 500 million.

Variable Cost
for each craft = $20x - 5x^{3/4} + 0.01x^2$

$$C(x) = 20x - 5x^{3/4} + 0.01x^2 + 500$$

$x \rightarrow$ number of aircraft produced while cost is in millions

As, $x(p) = 320 - 7.7p$ As $x(p) \rightarrow x$

$$x = 320 - 7.7p$$

$$7.7p = 320 - x$$

$$p = \frac{320 - x}{7.7}$$

① This is the demand function

~~Replacing p & x~~

$$x = \frac{320 - p}{7.7}$$

$$p(x) = \frac{320 - x}{7.7} \rightarrow \text{demand.}$$

$$R(u) = p(u) \cdot u$$

$$R(u) = \frac{320-u}{7.7} \cdot u \Rightarrow R(u) = \frac{320u - u^2}{7.7}$$

As, $p(u) = \frac{320-u}{7.7}$, this indicates the price at which people will buy, so demand price

$P(u) \rightarrow$ profit function

$$P(u) \rightarrow R(u) - C(u) \Rightarrow \frac{320u - u^2}{7.7} - 20u - 5u^{3/4} + 0.01u^2 + 500$$

$$P(u) = \frac{320u - u^2}{7.7} - 20u + 5u^{3/4} - 0.01u^2 + 500$$

Marginal Profit $P'(u)$

$$P'(u) = \frac{1}{7.7} (320 - 2u) - 20 + 5 \cdot \frac{3}{4} u^{1/4} - 0.02u$$

$$P'(12) = \frac{1}{7.7} (320 - 24) - 20 + \frac{15}{4(12)^{3/4}} - 0.02(12)$$

$$P'(12) = 38.44 + 20 + 2 - 0.24$$

$$P'(12) = 20.16$$

This interprets that, for selling 12 scripts we would have a profit of 20.16 million dollars.

Question 2 [10 Marks]

Check the continuity of $f(x) = \begin{cases} x \sin\left(\frac{|x|}{x}\right) & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ at $x = 0$.

We check the left hand limit at $x=0$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x \sin\left(\frac{|x|}{x}\right) !$$

By definition of mod,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x \sin\left(-\frac{x}{x}\right)$$

$$= \lim_{x \rightarrow 0^-} x \sin(-1)$$

$$\Rightarrow \sin(-1) \lim_{x \rightarrow 0^-} (x) \Rightarrow \sin(-1) \cdot 0 = 0$$

$$\therefore \boxed{\lim_{x \rightarrow 0^-} f(x) = 0} \quad \text{--- (1)}$$

Now checking the right hand limit.

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x \sin\left(\frac{|x|}{x}\right)$$

By definition of mod,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x \sin\left(\frac{x}{x}\right)$$

$$\lim_{x \rightarrow 0^+} x \sin(1) \Rightarrow \sin(1) \lim_{x \rightarrow 0^+} (x)$$

$$= \sin(1) \cdot 0 = 0$$

So,

$$\lim_{u \rightarrow 0^+} f(u) = 0 \quad \text{--- (II)}$$

So, limit exists at $u=0$

Also,

$$f(0) = 0 \quad \text{--- (III)}$$

By equation (I), (II) & (III), we get

$$\lim_{u \rightarrow 0^+} f(u) = \lim_{u \rightarrow 0^-} f(u) = f(0)$$

Hence,

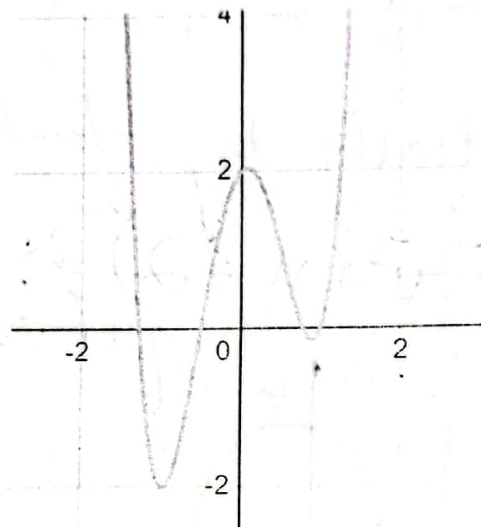
function $\{f(u)\}$ is continuous at $u=0$.

(10)

Question 3 [10 Marks]

Let $f(x) = 4x^4 - 7x^2 + x + 2$ be a function whose graph is given. Sketch the graph of the following:

- Compress the graph horizontally by a factor of 3 and followed by the reflection about x-axis.
- Compress the graph vertically by a factor of 2 and followed by the reflection about y-axis.

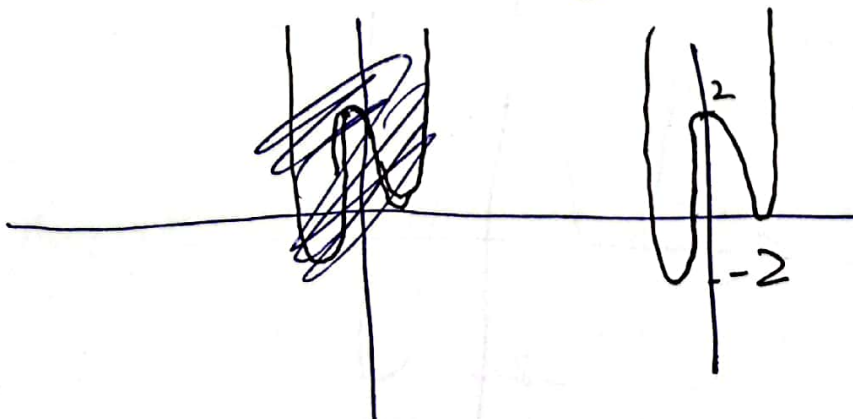


a) $\Rightarrow$ Compress horizontally by 3, so we get the equation as.

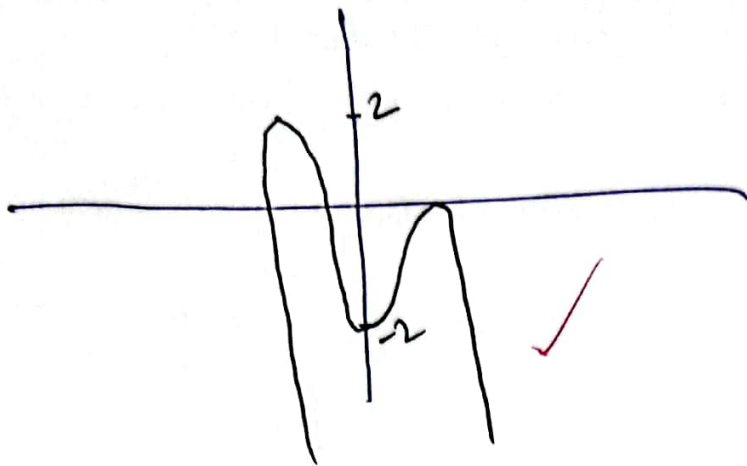
$$\Rightarrow 4(3u)^4 - 7(3u)^2 + 3u + 2$$

$$\Rightarrow 4 \cdot 81(u)^4 - 7 \cdot 9(u)^2 + 3u + 2$$

$$\Rightarrow 324u^4 - 63u^2 + 3u + 2$$

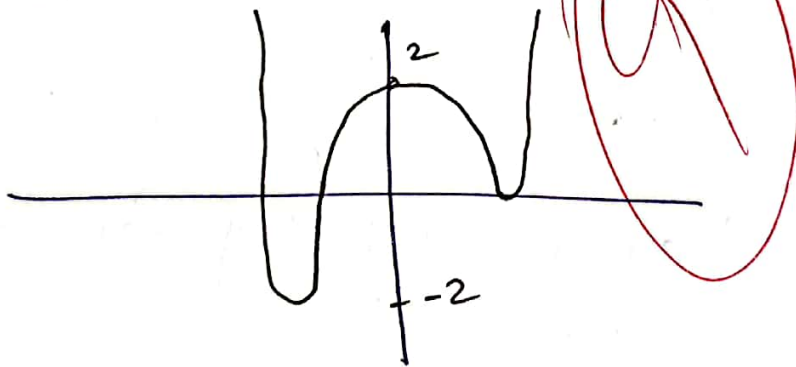


Also, Reflection obt. x -axis,
 $\text{Eqn} \Rightarrow -32u^4 + 63u^2 - 3u - 2$



(b). Compress Vertically by factor of 2.

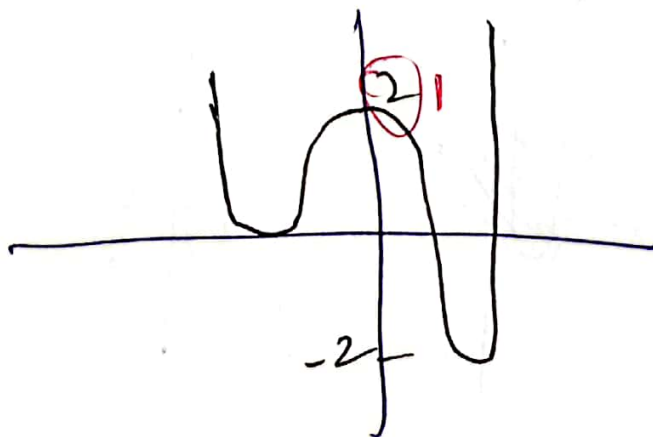
① $\Rightarrow \frac{1}{2}(4u^4 - 7u^2 + u + 2) \Rightarrow 2u^4 - \frac{7}{2}u^2 + \frac{u}{2} + 1$



$\frac{1}{2}f(u)$

② Reflection obt. y -axis.

$\text{Eqn} \Rightarrow 2u^4 - \frac{7}{2}u^2 - \frac{u}{2} + 1$



Question 4 [10 Marks]

Find the limit if it exists. If the limit does not exist, explain why?

a) $\lim_{x \rightarrow 0^+} \sqrt{x} \left(1 + \sin^2 \left(\frac{2\pi}{x} \right) \right)$.

b) $\lim_{x \rightarrow 0^-} \sqrt{x} \left(1 + \sin^2 \left(\frac{2\pi}{x} \right) \right)$.

c) $\lim_{x \rightarrow 0} \sqrt{x} \left(1 + \sin^2 \left(\frac{2\pi}{x} \right) \right)$.

a) $\lim_{u \rightarrow 0^+} \sqrt{u} \left(1 + \sin^2 \left(\frac{2\pi}{u} \right) \right)$
 $\rightarrow \lim_{u \rightarrow 0^+} \sqrt{u} + \sqrt{u} \sin^2 \left(\frac{2\pi}{u} \right)$
 $\rightarrow \lim_{u \rightarrow 0^+} \sqrt{u} + \lim_{u \rightarrow 0^+} \sqrt{u} \sin^2 \left(\frac{2\pi}{u} \right)$
 $\rightarrow 0 + \lim_{u \rightarrow 0^+} \sqrt{u} \sin^2 \left(\frac{2\pi}{u} \right) \quad \text{--- (1)}$

As $-1 \leq \sin^2 \left(\frac{2\pi}{u} \right) \leq 1$
 $-\sqrt{u} \leq \sqrt{u} \sin^2 \left(\frac{2\pi}{u} \right) \leq \sqrt{u}$

By Squeeze Theorem.

$$\lim_{u \rightarrow 0^+} -\sqrt{u} \leq \lim_{u \rightarrow 0^+} \sqrt{u} \sin^2 \left(\frac{2\pi}{u} \right) \leq \lim_{u \rightarrow 0^+} \sqrt{u}$$

$$g(u) \leq f(u) \leq h(u)$$

$$\lim_{u \rightarrow 0^+} g(u) = 0 \quad \& \quad \lim_{u \rightarrow 0^+} h(u) = 0, \text{ So}$$

By Squeeze Theorem,

$$\lim_{u \rightarrow 0^+} f(u) = 0 \Rightarrow \lim_{u \rightarrow 0^+} \sqrt{u} \sin^2 \left(\frac{2\pi}{u} \right) = 0$$

So, eq-① becomes

$$S_0 = 0 + 0 = 0$$

$$\lim_{n \rightarrow 0^+} \sqrt{n} \left(1 + \sin^2\left(\frac{2\pi}{n}\right) \right) = 0,$$

Thus, This limit exists.

⑥. $\lim_{n \rightarrow 0^-} \sqrt{n} \left(1 + \sin^2\left(\frac{2\pi}{n}\right) \right)$

$$\lim_{n \rightarrow 0^-} \sqrt{n} + \sqrt{n} \sin^2\left(\frac{2\pi}{n}\right)$$

$$= \lim_{n \rightarrow 0^-} \sqrt{n} + \lim_{n \rightarrow 0^-} \sqrt{n} \sin^2\left(\frac{2\pi}{n}\right)$$

As, we cannot approach $\sqrt{n}$ by $n \rightarrow 0^-$, so this limit is not defined.

$$\lim_{n \rightarrow 0^-} \sqrt{n} \left(1 + \sin^2\left(\frac{2\pi}{n}\right) \right) \Rightarrow \text{Does not exist.}$$

⑦

As. $\lim_{n \rightarrow 0^+} \sqrt{n} \left(1 + \sin^2\left(\frac{2\pi}{n}\right) \right) = 0$ &

$$\lim_{n \rightarrow 0^-} \sqrt{n} \left(1 + \sin^2\left(\frac{2\pi}{n}\right) \right) \Rightarrow \text{Does not exist.}$$

Since, L.H.L $\neq$ R.H.L

So,

$$\lim_{n \rightarrow 0} \sqrt{n} \left(1 + \sin^2\left(\frac{2\pi}{n}\right) \right) \Rightarrow \text{Does not exist.}$$

Question 5 [10 Marks]

Define an Asymptote? For the function $f(x) = \frac{x^2 - x}{x^2 + x - 12}$, find all possible asymptotes, also find the x-coordinate of the point of intersection of the graph of the function with its asymptote.

Asymptote:

An asymptote is a line, that a graph seems to follow very closely, but the graph never crosses it in case of vertical line. The graph tends to go along the line very closely that we assume that its distance with that line is zero.

$$f(x) = \frac{x^2 - x}{x^2 + x - 12}$$

For H.A:

As, Powers of Numerator i.e 2 & Powers of Denominator i.e 2 are equal. So, H.A $\Rightarrow$ $y = 1$

For Oblique Asymptote:

Since, There exists an H.A so oblique asymptote doesn't exist.

$$f(u) = \frac{u^2 - u}{u^2 + u - 12}$$

$$f(u) = \frac{u(u-1)}{(u-3)(u+4)}$$

, Since, at $u=3$ & $u=-4$, there is a non-removable discontinuity

So, for V.A

$$(u-3)(u+4) = 0$$

$$\boxed{u=3} \text{ \& } \boxed{u=-4}$$

we have two Vertical Asymptotes
Since, The graph can touch H.A, so, finding their intersection.

$$\frac{u(u-1)}{(u-3)(u+4)} = 1$$

$$\cancel{u^2} - u = \cancel{u^2} + u - 12$$

$$\cancel{+2u} = \cancel{-12}$$

$$\boxed{u=6} \text{ u-coordinate}$$

So, The graph touches the H.A at (6, 1)

